ASSIGNMENT -3 -CO3

NAME:	**M.MANASA**
REG NO:	**192425189**
COURSE CODE:	**ITA0329**
COURSE:	**MOBILE COMPUTING**

Q1 – Wireless Network Architecture and Mobility Analysis

Part A – Introduction to Wireless Networks

A wireless network connects devices without using physical cables. It uses radio waves to transfer data between mobile devices and network infrastructure.

1. Characteristics of Wireless Networks

- Mobility: Users can move while remaining connected to the network.
- Limited bandwidth: Wireless channels usually have less capacity than wired links.
- Variable signal strength: Distance, obstacles and interference can affect signal quality.
- Shared medium: Multiple devices use the same wireless channel.
- Handoff: A mobile device may change from one access point or base station to another.
- Security: Wireless communication needs authentication and encryption.

2. Main Components and Their Roles

Component	Role
Mobile node	A moving device such as a smartphone, laptop or ambulance terminal.
Wireless access point / Base station	Provides wireless access and connects mobile nodes to the network.
Communication link	Radio connection used to exchange packets.
Router / Gateway	Forwards packets between networks.
Network server	Provides services, data or applications to users.

3. Wireless Network Architecture

Mobile Node 1)))

Wireless link

Mobile Node 2))) ---> Access Point / Base Station ---> Router/Gateway ---> Internet/Server

Mobile Node 3)))

In this architecture, mobile nodes communicate with an access point or base station. The access point forwards their packets to the router and then to the required server or destination.

4. Communication Constraints

Constraint	**Effect on communication**
Limited bandwidth	Reduces data rate when many devices share the channel.
Weak connectivity	May cause disconnection or retransmission.
Mobility	Can change the point of attachment and interrupt a session.
Interference	Can increase errors and packet loss.

Part B – Wireless Mobility Scenario

Scenario: A user is moving from the coverage area of Access Point A to the coverage area of Access Point B while making a video call.

1. The mobile device is initially connected to Access Point A.
2. As the user moves away, the signal from A becomes weak.
3. The device detects Access Point B and checks its signal quality.
4. If B provides a suitable signal, the device performs handoff and connects to B.
5. Communication continues through B. During the handoff, a short delay or packet loss may occur.

Successful and Interrupted Communication

Case	**Behaviour**
Successful handoff	The new access point is detected and connection is transferred quickly. Most packets reach the destination.

Interrupted handoff	The new network is not found or registration takes too long. Packets may be delayed or lost.
High mobility	The device changes coverage areas frequently, causing repeated handoffs and more interruptions.

Performance measures: packet delivery ratio = (packets received / packets sent) × 100. End-to-end delay is the time taken by a packet to reach its destination. Packet loss is the number of packets that do not reach the destination.

Conclusion: Wireless networks support communication while users move, but bandwidth limits, signal changes and handoff delays can affect performance. Good coverage and fast handoff improve communication.

Q2 – Mobile IP and Mobility Management

Mobile IP allows a mobile node to move from one network to another while keeping its permanent home IP address. It helps maintain ongoing communication even when the node changes its network attachment.

1. Mobile IP Components

Component	Meaning / Function
Mobile Node (MN)	The device that moves between networks.
Home Agent (HA)	A router on the home network that stores the mobile node's current location and forwards packets.
Foreign Agent (FA)	A router in a visited network that assists the mobile node in traditional Mobile IPv4.
Correspondent Node (CN)	The device or server communicating with the mobile node.
Care-of Address (CoA)	The temporary address representing the mobile node's current location.

2. Mobile IP Architecture

Correspondent Node (CN)

| Packet addressed to MN's Home Address

v

Home Network: Home Agent (HA)

| Encapsulates packet and tunnels it

v

Visited Network: Foreign Agent (FA) / Care-of Address

|

v

Mobile Node (MN)

The mobile node keeps its home address. When it enters a visited network, it obtains a care-of address and registers it with the Home Agent.

3. Registration and Location Updating

6. The mobile node detects that it is in a visited network.
7. It discovers a Foreign Agent or obtains a care-of address.
8. The mobile node sends a registration request containing its home address and care-of address to the Home Agent.
9. The Home Agent authenticates and checks the request.
10. If accepted, the Home Agent stores the binding between the home address and care-of address.
11. The Home Agent sends a registration reply.

4. Packet Delivery and Tunnelling

12. The Correspondent Node sends a packet to the mobile node's permanent home address.
13. The packet reaches the home network and is intercepted by the Home Agent.
14. The Home Agent encapsulates the original packet inside another packet and sends it through a tunnel to the care-of address.
15. The tunnel endpoint decapsulates the packet and delivers the original packet to the mobile node.

5. Important Mobile IP Challenges

Challenge	**Explanation / Possible improvement**
Registration delay	Registration takes time after movement. Reduce unnecessary registrations and use efficient agent discovery.
Tunnelling overhead	Extra headers increase packet size. Use efficient encapsulation.
Triangular routing	Packets may travel CN → HA → MN instead of directly to MN. Route optimisation can reduce the indirect path where supported.
Packet loss	Packets may be lost during handoff. Buffer packets or retransmit when appropriate.
Security	False registration can redirect traffic. Authenticate registration messages and protect mobility information.
Scalability	Many mobile nodes create signalling load. Use efficient registration and forwarding methods.

6. Joint Answer: Frequent Network Changes During an Active Session

Frequent movement may cause repeated agent discovery, registration delay, changing care-of addresses, packet loss and additional tunnelling overhead. The session may experience delay or temporary interruption. These problems can be reduced by fast agent discovery, timely registration, buffering packets during handoff, selecting a strong target network, authenticating registration messages and reducing unnecessary signalling.

Conclusion: Mobile IP supports location-independent communication by registering a care-of address and tunnelling packets through the Home Agent. Its main concerns are delay, overhead, packet loss, security and scalability.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks

1. Cellular Network

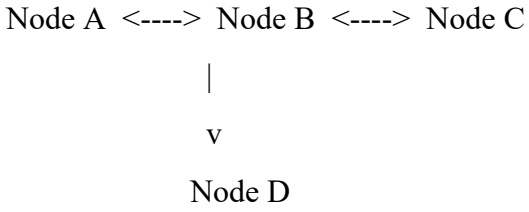
A cellular network is an infrastructure-based wireless network. A geographical area is divided into cells. Each cell is served by a base station. Mobile devices communicate through the base station, which connects them to the wider network.

Mobile Device))) Base Station / Cell Tower ---> Core Network ---> Internet

- Base stations manage wireless access and support handoff between cells.
- Communication is mainly infrastructure-assisted and centrally managed.
- It provides wide-area coverage when the required infrastructure is available.

2. Ad Hoc Wireless Network

An ad hoc wireless network is an infrastructure-less network formed by mobile nodes. It does not require a fixed base station or access point. Nodes communicate directly when they are in range, or through other nodes using multi-hop communication.



- Each node can work as both a host and a router.
- The topology changes when nodes move or links fail.
- It can be deployed quickly in places where fixed infrastructure is unavailable.

3. Comparison

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Uses base stations and fixed infrastructure.	Does not require fixed infrastructure.
Communication	Usually through a base station.	Direct or multi-hop between mobile nodes.
Routing	Infrastructure-assisted routing.	Distributed routing between nodes.
Mobility	Uses cellular mobility and handoff management.	Routes must adapt to moving nodes.

Topology	Planned cell structure.	Dynamic topology.
Scalability	Can support many users with planned capacity.	Depends on node density, wireless links and routing overhead.
Handoff	Handoff between cells/base stations.	Routes or neighbor links are updated as nodes move.
Failure handling	Base-station or core-network failure can affect service.	Alternative peer paths may be used if available.
Applications	Mobile voice, broadband and public cellular service.	Disaster recovery, military, temporary and field networks.

4. Joint Engineering Decision – Disaster Environment

For an emergency or disaster area where fixed infrastructure may be unavailable, an ad hoc wireless network is a suitable choice. Rescue teams, police and ambulances can form a temporary network and forward messages through nearby nodes. Multi-hop communication can extend coverage beyond the range of one device. If some nodes or links fail, the routing protocol can search for another available path.

Limitation: An ad hoc network may have limited bandwidth, changing routes and battery constraints. If cellular infrastructure is available and reliable, it can provide wider coverage and managed service.

Conclusion: Cellular networks are suitable when infrastructure and broad coverage are available. Ad hoc networks are useful for rapid communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements

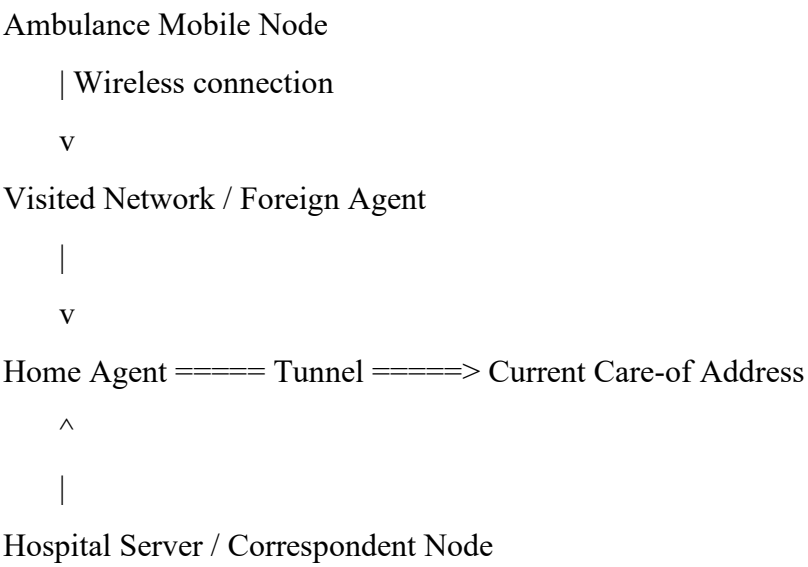
Selected Application: Emergency Ambulance Communication

Mobile IP can be used in an emergency healthcare system where ambulances move between different network areas but must remain connected to hospitals.

1. Network Entities

- Mobile Node: Ambulance communication device or onboard computer.
- Home Agent: Router that maintains the ambulance's home-address-to-care-of-address binding.
- Foreign Agent / visited network: Network currently serving the ambulance.
- Correspondent Node: Hospital server or emergency control centre.

2. Application Architecture



The ambulance sends patient details, location and emergency updates to the hospital. When it enters another network, it obtains a new care-of address and registers it with the Home Agent. Packets can then be tunnelled to its current location.

3. Communication Flow

16. The ambulance device connects to the available wireless network.
17. It discovers the visited network and obtains a care-of address.
18. It registers the care-of address with the Home Agent.
19. The hospital server sends information to the ambulance's home address.
20. The Home Agent tunnels packets to the current care-of address.
21. The ambulance receives the information and continues sending updates.

4. Routing and Performance Requirements

Requirement	Why it is important
Low delay	Emergency instructions and patient information must arrive quickly.
High packet delivery	Important medical data should reach the hospital correctly.
Continuous connectivity	Communication should continue while the ambulance moves.
Fast handoff	Network changes should cause minimum interruption.
Security	Patient and emergency information must be protected.
Reliability	The system should detect failures and use an available connection or route.

5. Suitability of Mobile IP

Mobile IP is suitable because it allows the ambulance to keep its permanent home address while moving between networks. It supports communication with the hospital without requiring the correspondent to know the ambulance's changing location. However, registration delay, packet loss and tunnelling overhead must be managed.

Conclusion: Mobile IP can support mobile healthcare and ambulance communication by maintaining a binding to the current location and forwarding packets through a tunnel.

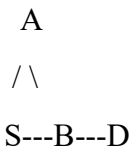
Q5 – Proactive Routing Protocols in Mobile Networks

A proactive routing protocol maintains routes to destinations before data is sent. Nodes periodically exchange routing information and update their routing tables. Therefore, a route may already be available when an application needs to send a packet.

1. DSDV Protocol

Destination-Sequenced Distance Vector (DSDV) is a proactive routing protocol. It uses destination sequence numbers to identify fresh routes and reduce routing loops. Each routing-table entry contains destination, next hop, hop count and sequence number.

2. Sample Network Topology



Example: Node S can reach destination D through B. If the S–B link fails, the network exchanges updated routing information and searches for a valid route through another neighbor if one exists.

3. DSDV Route Table Example

Node	Destination	Next Hop	Hop Count	Sequence No.	Status
S	D	B	2	20	VALID
B	D	D	1	20	VALID
A	D	B	2	20	VALID

After a topology change, a node receives a newer sequence number or a route with a better metric. It updates the routing table with the fresher or better route. Invalid or expired entries are removed.

4. Simplified DSDV Route Update Pseudocode

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT: Destination, NextHop, HopCount, SequenceNumber
OUTPUT: Updated routing table
BEGIN
  Destination ← ReceivedRoute.Destination
  NewHopCount ← ReceivedRoute.HopCount + 1
  IF Destination is not in RouteTable THEN
```

```
    ADD the received route
  ELSE IF Received SequenceNumber is newer THEN
    REPLACE the old route
  ELSE IF sequence numbers are equal AND NewHopCount is smaller THEN
    REPLACE the old route
  END IF
  RETURN RouteTable
END
```

5. Performance Analysis

Parameter	**DSDV behaviour**
Route availability	Routes are maintained before data transmission.
Initial route delay	Usually low because routes are already present.
Control overhead	Periodic updates consume bandwidth.
Convergence	After topology changes, nodes need time to receive and process updates.
Mobility	Frequent movement causes more routing-table changes and updates.
Network size	Larger networks can increase the amount of routing information and control traffic.

Conclusion: DSDV is useful when low route-discovery delay is important and the network topology is reasonably stable. Its main disadvantage is periodic control overhead, especially when nodes move frequently.

Q6 – Reactive Routing and Protocol Comparison

A reactive routing protocol discovers a route only when a source node needs to send data and no valid route is available. This is called on-demand routing.

1. AODV Protocol

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. It discovers routes when required and maintains active routes while they are being used.

2. AODV Control Messages

Message	Purpose
RREQ – Route Request	Broadcast by the source to search for a route to the destination.
RREP – Route Reply	Sent back when the destination or a node with a valid route is found.
RERR – Route Error	Informs affected nodes that a route or link has failed.

3. AODV Route Discovery

22. Source S checks its routing table for a valid route to destination D.
23. If no route exists, S broadcasts a Route Request (RREQ) to its neighbors.
24. Intermediate nodes record a reverse route to S and forward the RREQ if they have not already processed it.
25. When D or an intermediate node with a valid route receives the request, it sends a Route Reply (RREP).
26. The RREP travels back to S and establishes the forward route.
27. S sends data using the discovered route.

4. Route Maintenance

28. Nodes monitor links on an active route.
29. If a link fails, the affected route is marked invalid.
30. If an alternate valid route exists, it is selected.
31. Otherwise, a Route Error (RERR) is sent to the source.
32. If communication is still required, the source starts a new route discovery.

5. AODV Route Discovery Pseudocode

```
ALGORITHM ReactiveRouteDiscovery(Source, Destination)
INPUT: Source and Destination
OUTPUT: Route or ROUTE_NOT_FOUND
BEGIN
  IF a valid route exists THEN RETURN the route
  Create a new Route Request (RREQ)
  Broadcast RREQ to neighbors
  WAIT for RREP or route failure
  IF RREP is received THEN
     Establish reverse and forward paths
     Mark route as VALID
     RETURN the route
  ELSE
     RETURN ROUTE_NOT_FOUND
  END IF
END
```

6. DSDV versus AODV

Feature	**DSDV (Proactive)**	**AODV (Reactive)**
Route discovery	Routes maintained continuously.	Route found when required.
Routing table	Regularly maintained with destination information.	Maintains routes needed for active communication.
Control overhead	Periodic update messages.	Mainly route-request and route-maintenance messages.
Initial delay	Usually low if route is already available.	May be higher while route discovery occurs.
Mobility	Frequent topology changes trigger table updates.	Broken active routes trigger repair or rediscovery.
Infrequent traffic	May waste bandwidth maintaining unused routes.	Can be efficient because routes are discovered on demand.
Example use	Networks needing ready routes	Networks with intermittent

	and frequent communication.	traffic and changing topology.

7. Joint Question – Highly Dynamic Network

For a highly dynamic mobile network with frequent topology changes, AODV is generally a suitable choice when traffic is intermittent or bandwidth is limited. It discovers routes only when needed, so it avoids continuously updating every route. When an active link fails, it uses route maintenance and can start a new discovery.

However, AODV has route-discovery delay when a route is not already known. If the application sends frequent traffic and requires very low initial delay, DSDV may be preferred because routes are maintained in advance. The final choice depends on traffic rate, mobility, bandwidth and delay requirements.

Conclusion: AODV is often suitable for dynamic networks with on-demand communication, while DSDV is useful when immediate route availability is more important than periodic control overhead.

Q7 – Design a Mobile Emergency Communication Network

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical staff move continuously. Fixed communication infrastructure may be unavailable. The network must provide reliable communication with minimum delay.

Stage 1 – Independent Design Proposals

Student A – Architecture and Mobility Proposal

- Select an ad hoc wireless network as the primary local communication architecture because fixed infrastructure may be unavailable.
- Treat ambulances, police vehicles and rescue-team devices as mobile nodes.
- Use multi-hop wireless links so nodes can forward messages for other nodes.
- Where a gateway or external network is available, use Mobile IP to maintain reachability as a mobile node changes its network attachment.
- Use fast detection of a new network, care-of address acquisition and registration with the Home Agent.

Student B – Routing and Failure Proposal

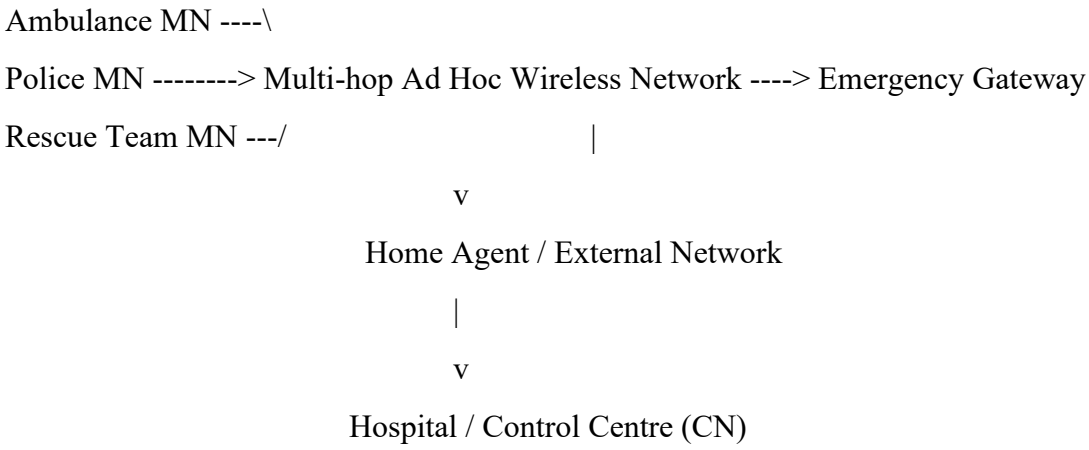
- Use AODV as the main routing protocol for the changing ad hoc network because it discovers routes on demand.
- Use RREQ for route discovery, RREP for route establishment and RERR for link-failure notification.
- Monitor active links and use an alternate route if available.
- If no route is available, start a new route discovery when communication is still required.
- Evaluate packet delivery ratio, end-to-end delay, routing overhead and route recovery time.

Stage 2 – Compare the Two Designs

Area	Student A proposal	Student B proposal	Joint decision
Architecture	Ad hoc local network with gateways where available.	Routing between mobile nodes.	Use a multi-hop ad hoc network.
Mobility	Mobile IP at the boundary with external	Route updates as nodes move.	Use Mobile IP for inter-network mobility

	networks.		and routing for local paths.
Routing	Focus on connectivity architecture.	AODV on demand.	Select AODV for dynamic local routes.
Failure handling	Detect loss of network attachment.	RERR and alternate route discovery.	Use both network reattachment and route repair.
Performance	Reduce handoff interruption.	Measure delay and delivery.	Measure delay, packet delivery and routing overhead.

Stage 3 – Joint Final Network Design



Local rescue teams exchange messages through the ad hoc network. When a gateway to an external network is available, the mobile node can use Mobile IP. The Mobile Node obtains a care-of address in the visited network and registers it with the Home Agent. The Home Agent tunnels packets addressed to the home address toward the care-of address.

Mobile IP Registration and Tunnelling

33. The mobile node detects a visited network.
34. It discovers an agent or obtains a care-of address.
35. It sends a registration request to the Home Agent.
36. The Home Agent authenticates the request and stores the home-address/care-of-address binding.
37. Packets from the Correspondent Node reach the Home Agent.
38. The Home Agent encapsulates and tunnels packets to the care-of address.
39. The tunnel endpoint decapsulates and delivers packets to the mobile node.

Selected Routing Protocol: AODV

AODV is selected for local ad hoc communication because the nodes move and the topology changes. AODV discovers routes only when data must be sent. RREQ searches for a path, RREP establishes the path, and RERR reports a broken route. This reduces continuous routing updates when communication is intermittent.

Failure Handling

Failure	**Action**
Mobile node moves	Update neighbor information and discover or repair a route.
Wireless link fails	Mark the route invalid; use an alternate route or send RERR.
No route exists	Start a new AODV route discovery.
Gateway becomes unavailable	Continue local ad hoc communication; use another available gateway if present.
Mobile IP registration fails	Retry registration and maintain local communication where possible.
Temporary disconnection	Buffer important data where supported and retransmit when connectivity returns.

Stage 4 – Performance Evaluation

Metric	**Meaning**	**How to evaluate**
End-to-end delay	Time taken for a packet to reach its destination.	Measure send time and receive time; calculate the difference.
Packet delivery ratio	Percentage of sent packets received.	(Packets received ÷ packets sent) × 100.
Routing overhead	Control traffic used for routing.	Count RREQ, RREP, RERR and other routing-control messages.
Route recovery time	Time needed to restore a route after failure.	Measure from link-failure detection until a usable route is available.
Handoff interruption	Time communication is	Measure the gap in successful

	interrupted during network change.	packet delivery during handoff.

Suggested Test Cases

Test	Condition	Expected observation
1	Small number of mobile nodes; low mobility.	Routes are relatively stable and delay is expected to be lower.
2	Increase the number of mobile nodes.	More route requests and control messages may occur.
3	Increase node speed.	Links change more often; route repairs may increase.
4	Break an active link.	AODV should report the failure and discover another route if possible.
5	Disconnect the gateway.	Local ad hoc communication may continue, but external communication may stop.

No measured simulation values are provided in the assignment question. The table above gives expected observations; actual numerical results should be added only after running a simulation or experiment.

Stage 5 – Cross-Defence

Why is AODV appropriate? It is demand-driven, avoids continuous route updates for unused destinations and provides route maintenance when an active link fails. This is useful when mobile nodes frequently change position.

How does Mobile IP maintain communication? It keeps the mobile node's permanent home address and registers the current care-of address with the Home Agent. The Home Agent tunnels packets to the mobile node's current location.

Stage 6 – Joint Evaluation

- The ad hoc architecture is suitable because it can operate without fixed base stations.
- Mobile IP is useful when a node must remain reachable while moving between different IP networks; it is not a replacement for ad hoc routing inside the local network.

- AODV is selected for dynamic, on-demand local routing.
- The design introduces routing-control messages and Mobile IP registration/tunnelling overhead.
- High mobility can increase delay, packet loss and route-repair frequency.
- When connectivity is lost, local communication may continue if peer links remain; external communication may be unavailable until a gateway or route returns.
- The solution may be unsuitable when node density is too low, wireless links are unreliable, bandwidth is extremely limited or very strict delay guarantees cannot be met.

Final Conclusion

Before selecting Mobile IP and a routing protocol, a network designer should analyse infrastructure availability, node mobility, network size, traffic rate, bandwidth, delay requirement, packet delivery, security, energy limits and failure conditions. Mobile IP manages movement between IP networks, while routing protocols find paths between nodes. Both should be selected according to the needs of the application.